CEBU INSTITUTE OF TECHNOLOGY
UNIVERSITY

IT342-G1
SYSTEMS INTEGRATION AND ARCHITECTURE 1

FUNCTIONAL REQUIREMENTS SPECIFICATION (FRS)

Project Title: User Registration and Authentication

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1. Introduction

1.1. Purpose

The purpose of this document is to provide a detailed description of the functional and non-functional requirements for the User Registration and Authentication system. This document is intended for project stakeholders, developers, and quality assurance testers to ensure a shared understanding of the system's core identity management features.

1.2. Scope

The system provides a secure gateway for users to access protected application resources. Its boundaries include:

- User Registration: Allowing new guests to create accounts by providing names, emails, and passwords.
- Authentication: Verifying credentials to grant system access.
- Exclusions: This system does not include password recovery (forgot password) or third-party OAuth (e.g., Google/Facebook login) unless specified in future versions.

1.3. Definitions, Acronyms, and Abbreviations

- User – A person who registers and accesses the system
- Authentication – Process of verifying a user's identity
- Registration – Creating a new user account
- Credentials – User login details such as username and password
- FRS – Functional Requirements Specification
- SRS – Software Requirements Specification

2. Overall Description

2.1. System Perspective

This system acts as the primary security layer and entry point for a larger application ecosystem. It interacts directly with a database to store and retrieve user credentials and integrates with the application's landing page to manage user traffic based on authentication status.

2.2. User Classes and Characteristics

1. **Guest User:** Individuals seeking access. They are expected to have basic web navigation skills and a valid email address.
2. **Authenticated User:** Individuals who have successfully registered. They require the ability to view their specific profile and securely terminate their session.

2.3. Operating Environment

1. **Client Side:** Modern web browsers (Chrome, Firefox, Safari, Edge).
2. **Server Side:** Web server capable of executing validation logic.
3. **Database:** A relational database management system (RDBMS) to host the User table.

2.4. Assumptions and Dependencies

1. **Assumption:** Users have access to a stable internet connection.
2. **Dependency:** The system depends on a functioning database to persist user data during the registration "Saves data" step.

3. System Features and Functional Requirements

3.1. Feature 1: User Registration

Description: Allows unauthenticated guests to create a new identity within the system.
Functional Requirements:

- The system shall provide a registration form requesting name, email, and password.
- The system shall validate that the email is not already in use.
- The system shall save the new user record to the database upon successful validation.

3.2. Feature 2: User Authentication (Login)

Description: Verifies the identity of returning users to grant them access to the "Home" display.
Functional Requirements:

- The system shall allow users to input their email (username) and password.
- The system shall verify the credentials against the stored database records.
- Upon successful authentication, the system shall redirect the user to the application's Home page.
- The system shall provide a Logout function to end the active session.

4. Non-Functional Requirements

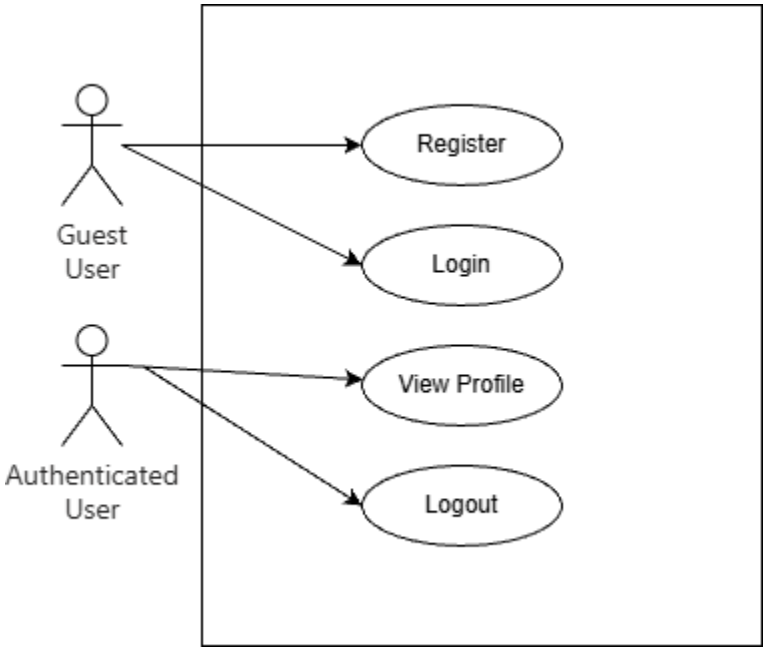
- **Security:** All passwords must be hashed before being saved to the database.
- **Performance:** Credential validation and login redirection should occur within less than 2 seconds.
- **Usability:** The "Create account form" must be mobile-responsive and include clear error messages for invalid inputs.
- **Availability:** The authentication service should be available 99.9% of the time to ensure users can always access the system.

5. System Models (Diagrams)

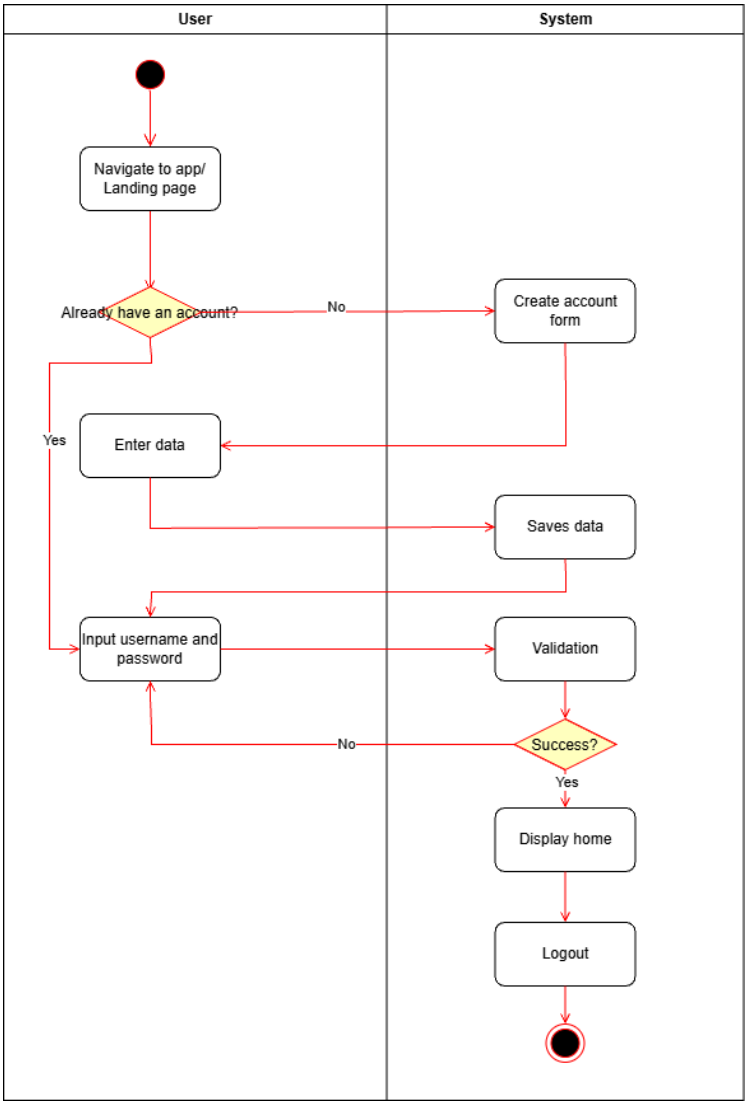
5.1. ERD

User	
PK	<u>**userID**</u>
	email username password

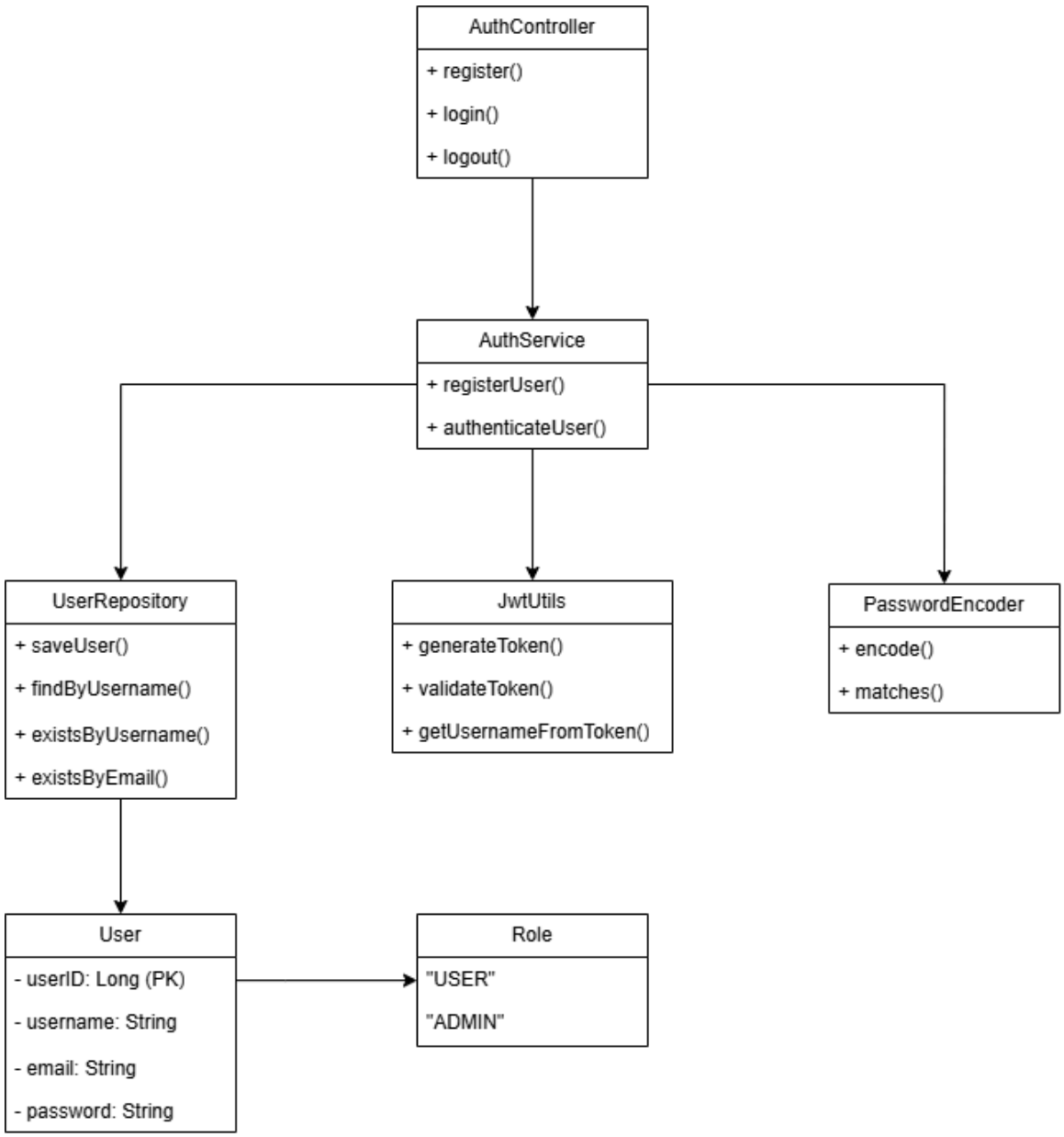
5.2. Use Case Diagram



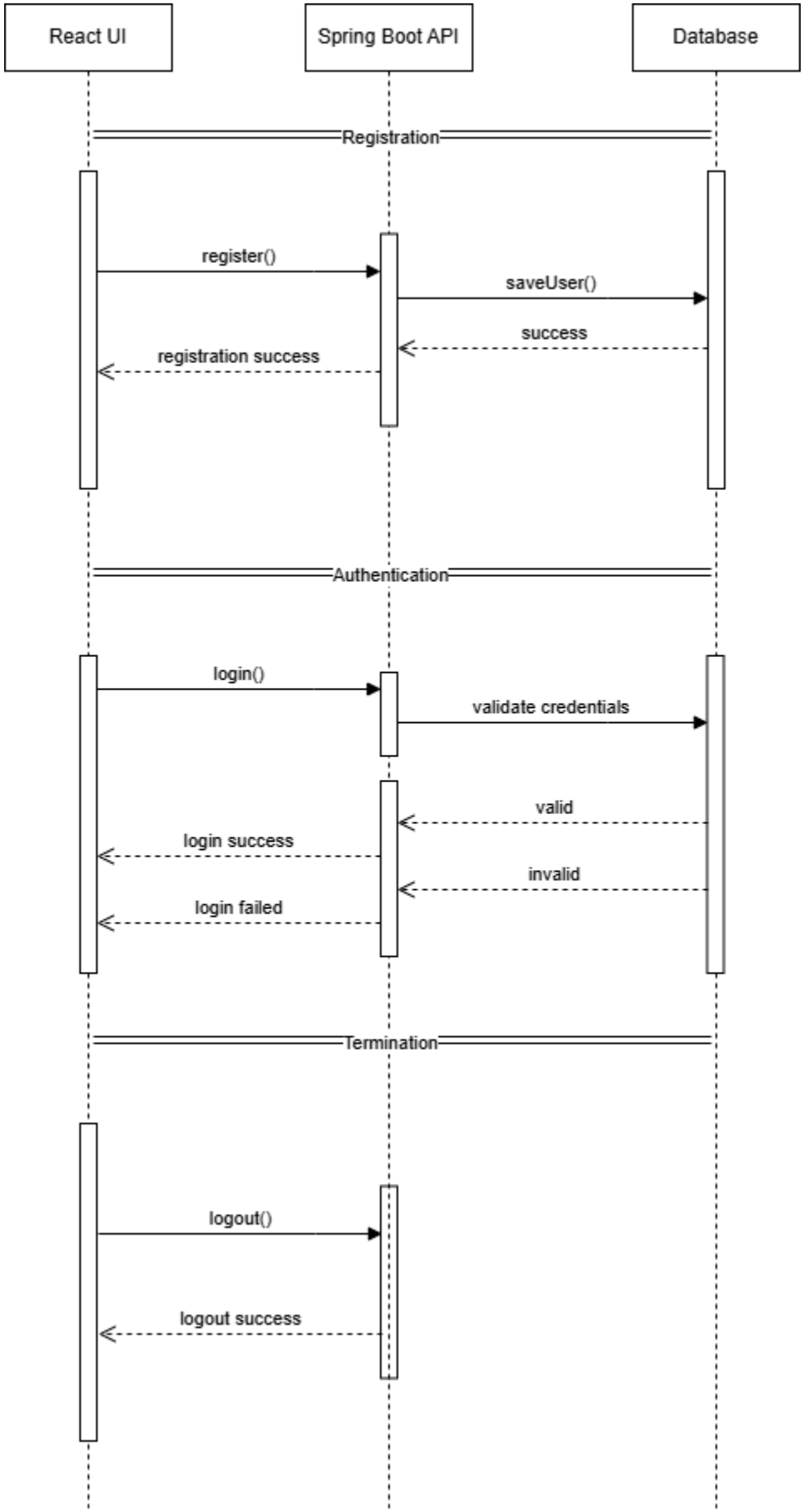
5.3. Activity Diagram



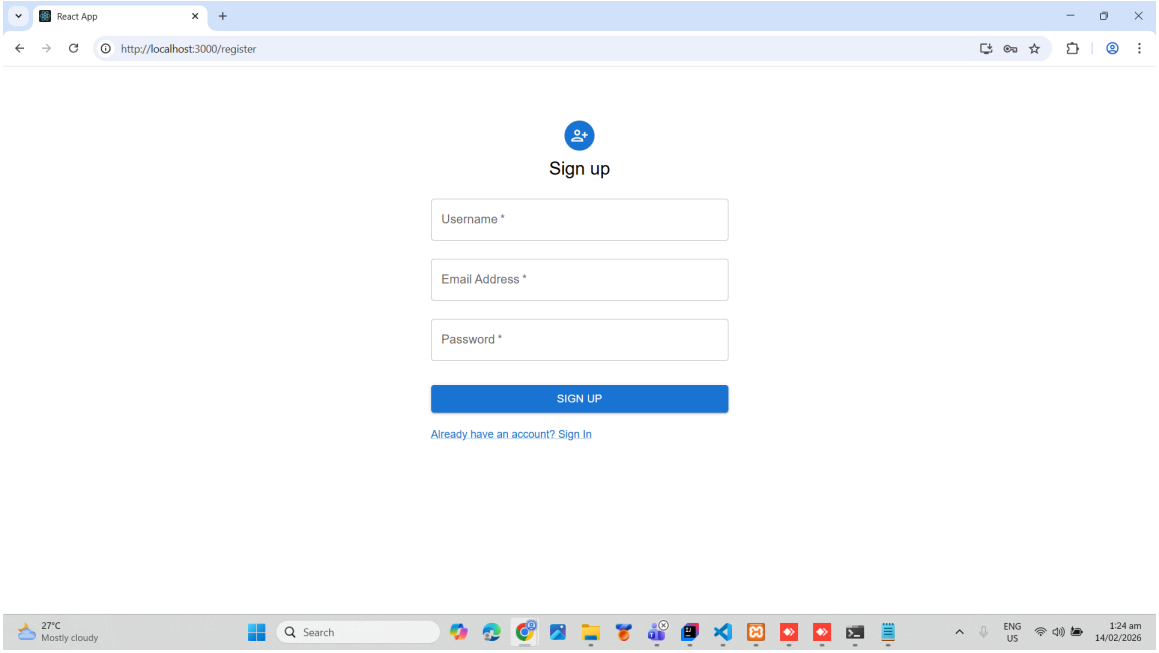
5.4. Class Diagram

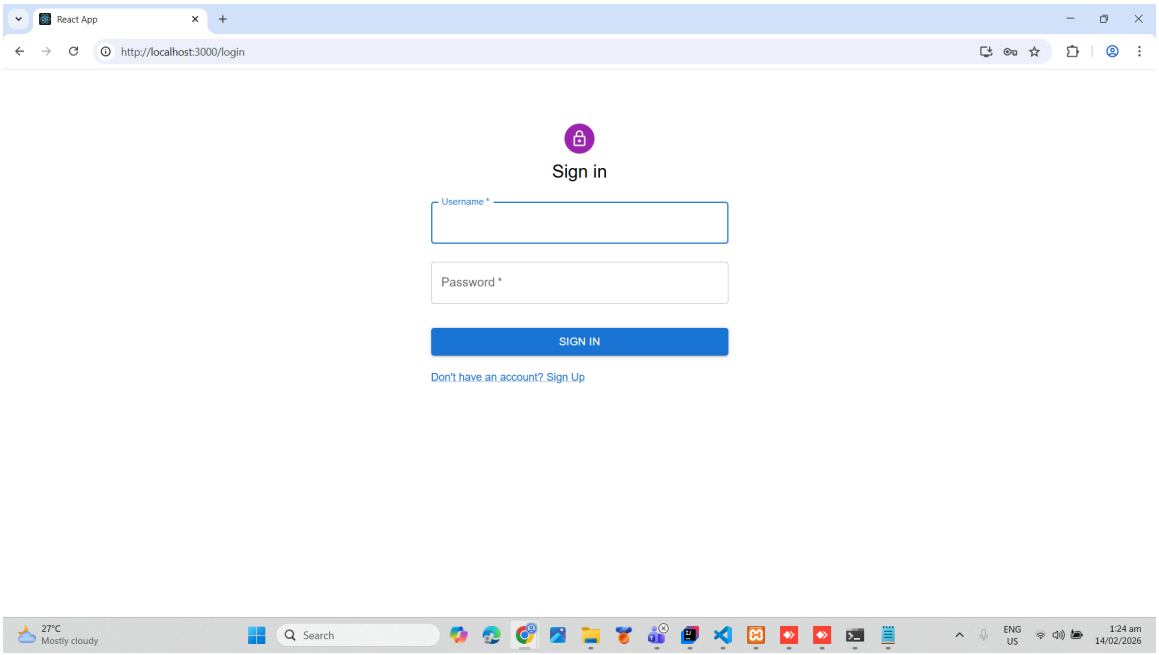


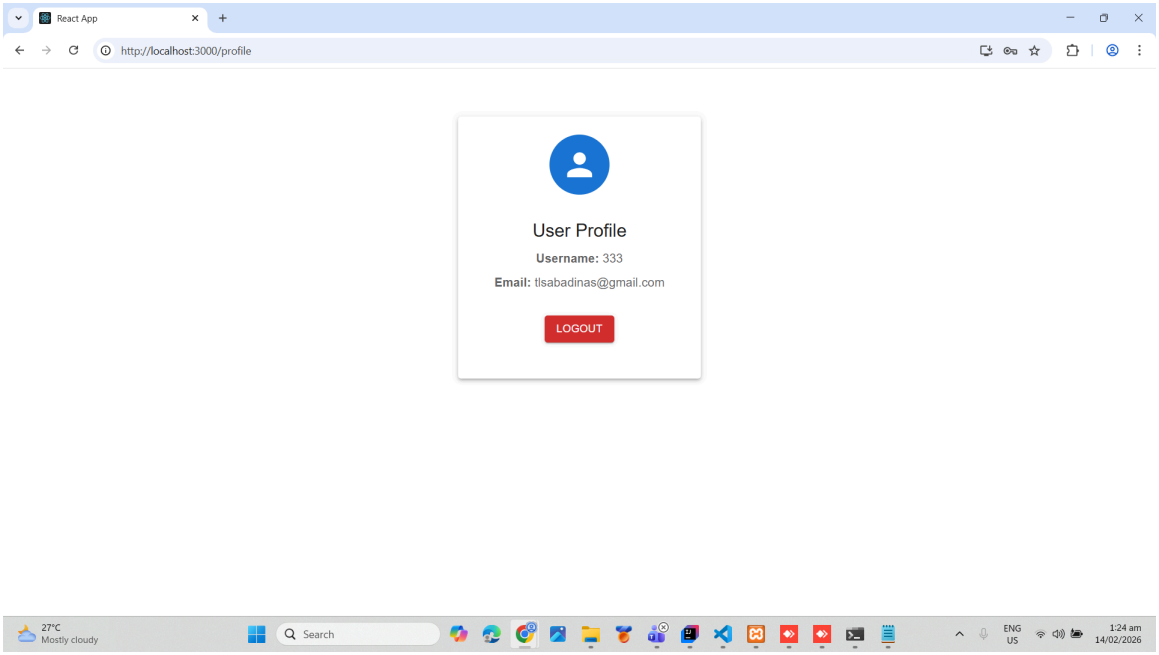
5.5. Sequence Diagram

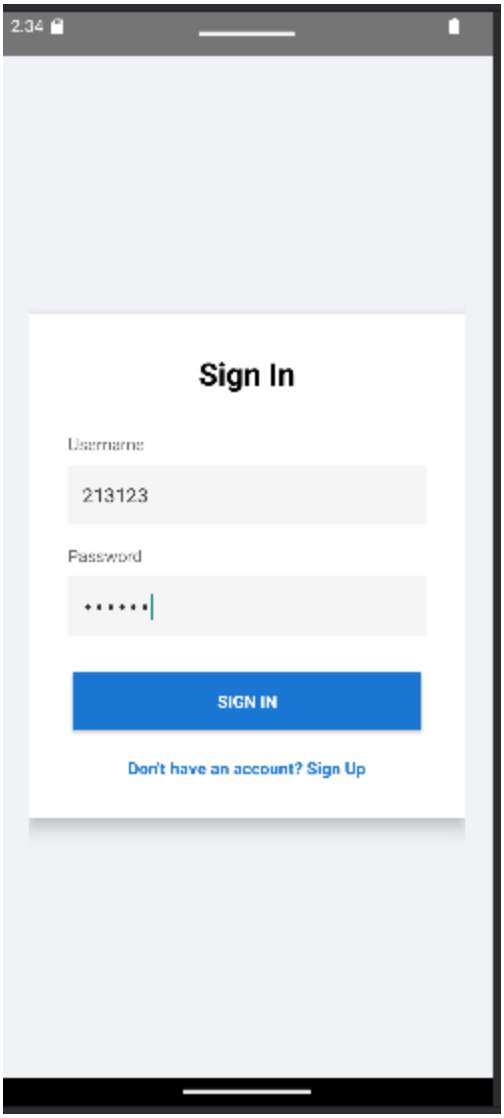


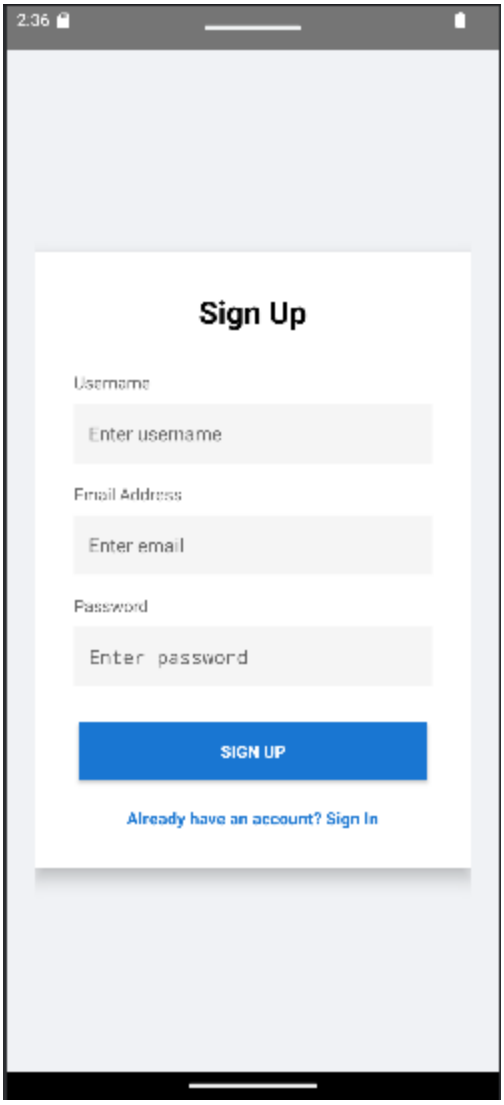
5.6. Screenshots of the Web UI


React App
http://localhost:3000/register
Sign up
Username *
Email Address *
Password *
SIGN UP
Already have an account? Sign In


React App
http://localhost:3000/login
Sign in
Username *
Password *
SIGN IN
Don't have an account? Sign Up


React App
http://localhost:3000/profile
User Profile
Username: 333
Email: tlsabadinas@gmail.com
LOGOUT
27°C
Mostly cloudy
Search
ENG
US
1:24 am
14/02/2026


2:34
Sign In
Username
213123
Password
SIGN IN
Don't have an account? Sign Up



6. Appendices

- Technological References: Developed using ReactJS for the UI, Spring Boot for the API, and MySQL for data storage.
- Tools: All diagrams were created using draw.io / diagrams.net.
- External Services: Assumes access to internet connectivity and correctly configured backend/database services.